

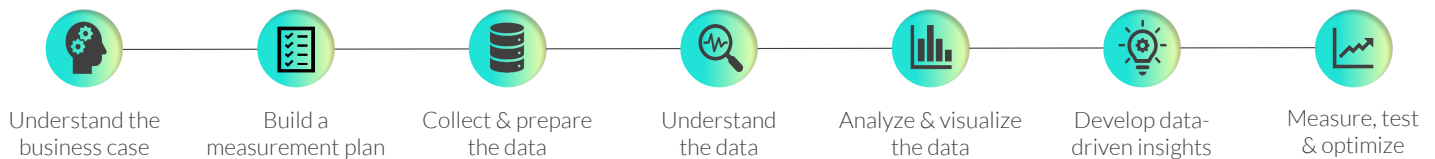
DATA ANALYSIS WORKFLOW

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DATA ANALYSIS WORKFLOW

WITHOUT A PROCESS, **YOU'RE FLYING BLIND**

This workflow will help you set clear expectations, define and measure key outcomes, improve accuracy and efficiency, and consistently deliver insightful, high-quality work



While there's no "right" or "wrong" workflow, **all analytics projects** should incorporate some combination of these steps

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UNDERSTAND THE BUSINESS CASE



Before you start thinking like an analyst, **think like a business owner.**

Understand which specific outcomes you are trying to impact, who the key stakeholders are and what motivates them, and how your analysis fits into the bigger picture.

This will help you align on requirements, project scope, and desired outcomes from day one.

KEY QUESTIONS:



- Which goals or business outcomes am I trying to impact?
- Who are the key stakeholders, and how will this help them?
- How does my work fit into the overall business strategy?

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BUILD A MEASUREMENT PLAN



Think of a measurement plan like a **roadmap for success.**

Measurement planning is about defining what a successful outcome looks like for the business, determining which KPIs align with that outcome, and identifying the data you need to capture, track and optimize those metrics.

If your goal is to deliver data-driven insights and outcomes, don't skip this step!

KEY QUESTIONS:



- What exactly does success look like for this analysis?
- Which specific metrics will help me quantify success?
- What data will I need to capture and track my KPIs?

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COLLECT & PREPARE THE DATA



An analysis is **only as strong as the data supporting it**.

Data prep tends to be one of the most challenging and time-consuming stages in the workflow, often involving a mix of quality assurance (QA), data profiling, feature engineering, and ETL automation (*more on that later!*).

This creates a strong foundation for your analysis, and ensures that you're working with clean, high-quality data.

KEY QUESTIONS:



- Where is my data stored, and how can I access it?
- Are there quality issues that need to be fixed?
- Will I need to model or transform the data before analysis?

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UNDERSTAND THE DATA



It may sound obvious, but it's important to **understand your data** before you begin your analysis.

You should have a crystal-clear understanding of the scope, granularity and composition of your source data, especially if your analysis requires some level of industry knowledge or domain expertise.

This will help you work efficiently and ensure that you have the exact data you need to support your analysis.

KEY QUESTIONS:



- What exactly does each record represent?
- Which fields are most relevant to my analysis?
- Are there important nuances or industry-specific metrics?

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ANALYZE & VISUALIZE THE DATA



Data visualization is about **bringing your data to life**.

Humans are poorly equipped to interpret and understand raw data; visualization is about creating clear patterns and visual cues to help us make sense of complex information.

This allows you to give your data a voice and create powerful, data-driven narratives.

KEY QUESTIONS:



- What type of data am I trying to analyze or visualize?
- What type of trend or pattern am I communicating?
- Who is the end user consuming this information?

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DEVELOP DATA-DRIVEN INSIGHTS



Remember that you don't get paid to analyze data; you get paid to **deliver insights & outcomes for the business**.

A strong insight not only tells a clear, data-driven story, but provides actionable recommendations designed to drive the key outcomes in your measurement plan.

This is one of the most important steps in the workflow, but one that many analysts fail to prioritize.

KEY QUESTIONS:



- What key takeaways are we able to derive from the data?
- How can we use these insights to impact real outcomes?
- How can I convince stakeholders to take action?

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MEASURE, TEST & OPTIMIZE



Understand the business case



Build a measurement plan



Collect & prepare the data



Understand the data



Analyze & visualize the data



Develop data-driven insights



Measure, test & optimize

You've completed your analysis, shared some strong data-driven insights, and delivered a set of actionable recommendations aligned with your measurement plan.

Great! Now what?

Once recommendations have been implemented, it's time to take credit. Track pre/post changes to your KPIs, quantify the impact of your work (*in dollars, if possible*), and seek opportunities to continue to test and optimize.

KEY QUESTIONS:



- Did my recommendations produce the desired outcomes?
- What value did this analysis generate for the business?
- Can this analysis help inform future tests or optimizations?